Investigation Topic: The Effect of Increasing Surface Area of the Blob on Damping of Pendulums

Clastify

1 Introduction

In physics class, pendulums are a useful demonstration of simple harmonic motion. However, from kids on a playground swing to geologists measuring seismic activities, pendulums also play a pivotal role in the real world. Applying physics to the real world requires careful consideration of details overlooked in theory, like damping caused by friction.

I wanted to further explore this topic because of an experiment done at school. In class, we learned that mass does not affect the oscillating period of a pendulum. In an ensuing lab, we tested the idea by using masses ranging from a coin to a tennis ball. Although the period measured did not differ greatly, there were variations beyond random error. I soon concluded that air resistance resulted in damping, but I could not determine the exact relationship from that lab because of the large uncertainty and the lack of controlled variables.

In this IA, I will investigate this phenomenon with more rigorous procedures. However, since this is an experiment done in a high school, the lack of sophisticated equipment will result in considerable uncertainty. Thus, I will improve the data-to-uncertainty ratio by using a light mass, boosting the effect of damping. This IA aims to answer the ***research question***: *To what extent can varying surface area (m^2) of a light mass affect the damping of a pendulum?*

2 Background Information

2.1 Simple Pendulum

A simple pendulum consists of a point mass on an inflexible, mass-less string with length L. In this system, only gravity and tension are acting forces, denoted by mg and T respectively. Since gravity always points toward the center of Earth, the horizontal component of the force is always against the displacement, generating a restoring force pointing at the equilibrium. Pendulums are a prime example of simple harmonic motion (SHM), so the acceleration of the pendulum is described as $a = -\omega^2 x$, where x is the displacement from the equilibrium and ω is the angular frequency.

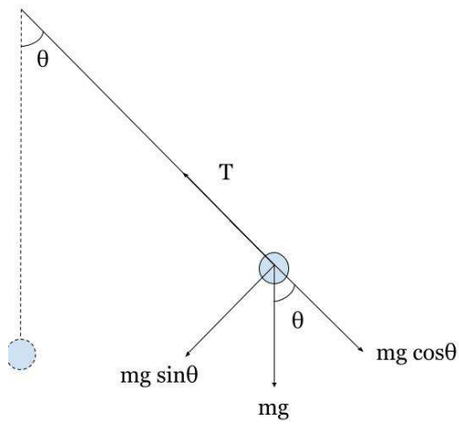


Figure 1: A free body diagram of a simple pendulum

2.2 Damping

Under ideal circumstances, a pendulum should stay in motion forever given the conservation of energy. However, damping dissipates the energy in the system, reducing the pendulum's motion. The type of damping discussed here is light damping: the bob will continue swinging but to a lesser amplitude in each oscillation. However, it is important to note that the period is unaffected by damping, as shown in Figure 2[1].

[1]Richmond, M. (n.d.). *Damped harmonic motion*. Apache2 Ubuntu Default Page: It works. http://spiff.rit.edu/classes/phys283/lectures/damped/damped.html

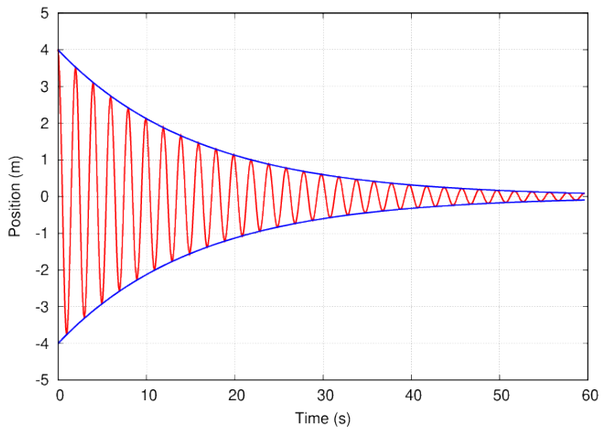


Figure 2: Example of light damping

2.3 Damping in Pendulum

In a pendulum, the main factor resulting in damping is drag. Since this experiment does not adopt a spherical bob and cannot guarantee streamlined flow, I cannot use Stoke's law. Instead, I will use the more generalized drag equation, which is defined as the following

$$F_D = -\frac{1}{2}\rho v^2 C_D A$$

where ρ is the density of the fluid (kg m^{-3}), v (m s^{-1}) is the moving object's velocity, C_D is the drag coefficient (unitless), and A is the cross-sectional area (m^2). The negative suggests that drag opposes the object's motion. F_D is a function of velocity because all other terms are constants, so we can rewrite drag as

$$F_D = -kv^2 \tag{1}$$

Recall that $\omega = \frac{2\pi}{T}$. Since the period is constant throughout, the force exerted by the pendulum is

$$F = ma = m(-\omega^2 x) = m(-\frac{4\pi^2}{T^2}x) \implies F = -mcx \tag{2}$$

where c is another constant.

Thus, the net force of the system is

$$\sum F = -kv^2 - mcx = ma \implies ma + kv^2 + mcx = 0 \tag{3}$$

Solving for x requires the equation to be in differential form,

$$m\frac{d^2x}{dt^2} + k(\frac{dx}{dt})^2 + mcx = 0 \implies \frac{d^2x}{dt^2} + \frac{k}{m}(\frac{dx}{dt})^2 + cx = 0 \tag{4}$$

I will borrow the results from Kharkonger & Mahato (2018)[2] because solving entails concepts of complex differentiation, which is far beyond the scope of this IA. They found

$$x(t) = x_{max}e^{-\frac{k}{2m}t}\cos(\omega t) \tag{5}$$

where x_{max} is the initial amplitude (m), k is the constant (kg m^{-1}) defined in 1, m is mass (kg), ω is the angular velocity (m^{-1}) and t is time (s).

With reference to Figure 2, the equation makes a lot of sense. The trigonometric portion describes the oscillating nature, and the exponential portion describes the damping effect.

[2]Kharkongor, D., & Mahato, M. C. (2018). Resonance oscillation of a damped driven simple pendulum. *European Journal of Physics, 39.* https://arxiv.org/pdf/1712.01032.pdf

3 Hypothesis

Clastify

This experiment aims to find the relationship between the surface area of the mass and the effect of dampin can be found within equation 5.

As defined above,

$$A \propto k \implies e^{-A} \propto e^{-\frac{k}{2m}t} \implies A \propto -\ln x(t)$$

Therefore, as the area increases, the amplitude decreases in a negative logarithmic relationship.

4 Experiment Design

4.1 Variables

The **independent variable** is the surface area of the mass. The different areas are 93.75 cm^2, 187.5 cm^2, 281.25 cm^2, and 375 cm^2. All dimensions are measured with a ruler with an error of ± 0.5 mm.

The **dependent variable** is the effect of damping, which can be measured through the loss of amplitude in each full oscillation. The experiment is recorded on video, so the loss is measured by the changes in pixel number, denoted by ΔP. The measurement is done by eye-balling, uncertainty is around ± 10 pixels.

The experiment uses several **controlled variables**.

- Initial Amplitude: Equation 5 contains amplitude, so the angle of release directly affects damping. Therefore, I placed a laser as a marker, and I released the mass when it blocked the beam.
- Shape of the Mass: The drag equation contains C_D, the drag coefficient, which varies depending on the shape of the object. Thus, all masses resembled a flat rectangular plate, which has a drag coefficient of 1.98[3].
- String Length: Maintaining the same length is important because $\omega = \sqrt{\frac{g}{L}}$, so changing the length will influence $x(t)$. The string length is 50 cm ± 0.5 cm.
- Bob Mass: This is the result of assuming air resistance is the dominant force. Increasing mass leads to increasing normal force and friction at the point of suspension. Although it is unclear at what mass will friction become dominant, I picked the lowest mass possible. The mass used was 38.43 g ± 0.01 g.
- Camera position: To reduce uncertainty and unnecessary parallax calculations, the experiment was done in one setting without ever moving the camera.

4.2 Apparatus

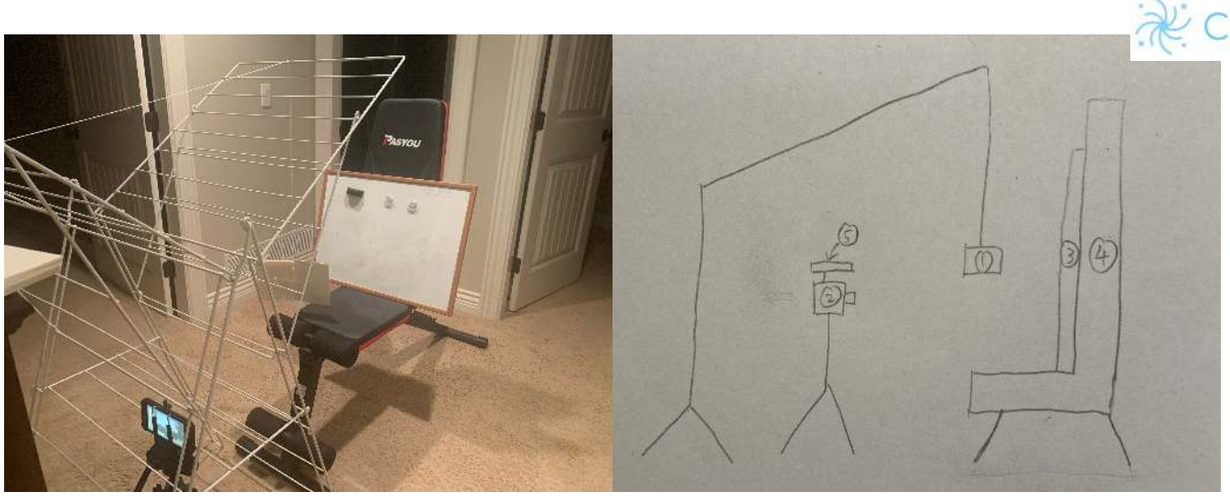
- Butcher twine (1)
- 12.5 × 7.5 cm chipboard sheets (24)
- High-speed camera (1)
- Meter ruler (± 0.1 cm)
- Tripod (1)
- Stapler (1) and stapler clips (16)
- Whiteboard (1)
- Drying Rack (1)
- Laser pen (1)
- Whiteboard Holder (1)
- Laptop (1)

A few things to note regarding the equipment. First, as mentioned above, I chose the lowest mass possible, which ended up being 24 sheets because it can divided into 1x24, 2x12, 3x8, and 4x6. Thus, the masses are 24 sheets together, 2 stacks of 12 sheets, etc. Second, I chose a drying rack over a ring stand because when the mass became 4 stacks of 6 sheets, the pendulum collided with the stand, so I had to get creative. Third, the whiteboard is marked with a ruler, but it acts as a contrasting background so that the bob's displacement can be better recognized.

[3] Aerospace, Mechanical & Mechatronic Engg. (2005). *Drag Coefficient.* Aerodynamics for Students. http://www-mdp.eng.cam.ac.uk/web/library/enginfo/aerothermal_dvd_only/aero/fprops/introvisc/node11.html

4.3 Design

Clastify



(a) Experiment setup (b) Simplified experiment setup

Figure 3: ① is the mass; ② is the camera, which is a phone in this case; ③ is the whiteboard; ④ is the whiteboard holder, which is a fitness bench here; ⑤ is the laser pen, which is underneath the phone in the real setup.

4.4 Methodology

1. Set up according to Figure 3 (b). Ensure ③ and ④ are perpendicular to the ground, and staple 4 clips onto the chip-board sheets.

2. Adjust the laser pen so that it is parallel with the camera, and point it towards the maximum amplitude. x_{max} in this experiment is 35 cm.

3. Connect the camera to a laptop. Use the pixel coordinate feature on the laptop to correspond pixels with displacement. For example, move the mass 5 cm and track how many pixels it has traveled. Maximize the accuracy by adjusting the camera so that the equilibrium and the max amplitude occupy both ends of the screen. The pixels-to-cm ratio is 13.4 pixels/cm.

4. Move the mass to the initial amplitude where it blocks the laser. Release without adding any lateral movements. End the recording when it completes 5 oscillations.

5. Repeat the trial 4 times. Then change the sheets to 2x12 and stabilize them with the stapler. Remove previous stapler clips.

6. Repeat step 5 with 3x8 and 4x6 setup, make sure to use 4 stapler clips throughout.

7. With the video, record the maximum amplitude of each complete oscillation in terms of its pixels away from equilibrium.

4.5 Risk Assessment

This experiment does not raise any notable ethical concerns. Still, I took several safety precautions. Firstly, I wore sunglasses during the experiment in case I accidentally looked at the laser pen. Secondly, the drying rack was secured with screws to prevent it from falling and causing injuries. Lastly, the experiment was completed away from crowds to avoid any accidents. The only environmental concern revolved around the disposal of the mass; instead of throwing it away, I recycled the sheets by using them as scratch papers.

5 Results

5.1 Table 1: Damping with Increasing Surface Area measured in Pixels

93.75 cm²					
number of oscillations	Displacement (pixels)				
	Trial 1	Trial 2	Trial 3	Trial 4	Average
1	347	336	350	352	346.25
2	242	269	282	288	270.25
3	218	220	215	223	219.00
4	185	181	179	182	181.75
5	134	161	157	155	151.75
187.5 cm²					
number of oscillations	Displacement (pixels)				
	Trial 1	Trial 2	Trial 3	Trial 4	Average
1	266	274	291	299	282.50
2	186	176	192	183	184.25
3	134	121	98	99	113.00
4	81	94	65	69	77.25
5	67	60	47	40	53.50
281.25 cm²					
number of oscillations	Displacement (pixels)				
	Trial 1	Trial 2	Trial 3	Trial 4	Average
1	254	243	260	250	251.75
2	157	134	138	148	144.25
3	77	80	83	84	81.00
4	41	40	43	36	40.00
5	28	30	23	27	27.00
375 cm²					
number of oscillations	Displacement (pixels)				
	Trial 1	Trial 2	Trial 3	Trial 4	Average
1	230	226	232	224	228.00
2	112	116	121	121	117.50
3	67	69	63	61	65.00
4	36	38	41	33	37.00
5	16	28	17	23	21.00

5.2 Table 2: Average Damping with Increasing Surface Area measured in Pixels

The table below only includes the average displacement of the four trials.

number of oscillations	Averaged Displacement (pixels)			
	93.75 cm²	187.5 cm²	281.25 cm²	375 cm²
1	346.25 ± 6	282.50 ± 13	251.75 ± 6	228.00 ± 3
2	270.25 ± 18	184.25 ± 6	144.25 ± 9	117.50 ± 4
3	219.00 ± 3	110.75 ± 18	81.00 ± 3	65.00 ± 3
4	181.75 ± 2	77.25 ± 11	40.00 ± 3	37.00 ± 3
5	151.75 ± 10	53.50 ± 11	27.00 ± 3	23.50 ± 5

I derived the uncertainty with standard deviation, which is defined as the following:

Clastify

$$\sigma = \sqrt{\frac{\sum_{i=1}^{k}(f_i x_i^2)}{n} - \bar{x}^2}$$

where f_i refers to the frequency, x_i is the value of the measured amplitude, n is the number of trials, and $\bar{x}$ is the mean measured amplitude.

Sample Calculation (1st oscillation in 375 cm³):

$$\begin{aligned}\sigma &= \sqrt{\frac{230^2 + 226^2 + 232^2 + 224^2}{4} - \left(\frac{230 + 226 + 232 + 224}{4}\right)^2} \\ &= \sqrt{51994 - 228^2} \\ &= \sqrt{10} \approx 3\end{aligned}$$

5.3 Table 3: Average Damping with Increasing Surface Area measured in cm

Pixels are not a useful unit to do any calculations, so I converted everything to centimetres. In step 3 of the methodology, I mentioned that I established a ratio between pixels and cm, which was 470 pixels: 35 cm. Thus, I multiplied each value in pixels with $\frac{35\ cm}{470\ pixels} = 0.0745\ cm/pixel$.

number of oscillations	Averaged Displacement (cm)			
	93.75 cm²	187.5 cm²	281.25 cm²	375 cm²
1	25.78 ± 0.45	21.04 ± 0.97	18.75 ± 0.45	16.98 ± 0.22
2	20.13 ± 1.34	13.72 ± 0.45	10.74 ± 0.67	8.75 ± 0.30
3	16.31 ± 0.22	8.25 ± 1.34	6.03 ± 0.22	4.84 ± 0.22
4	13.53 ± 0.15	5.75 ± 0.82	2.98 ± 0.22	2.76 ± 0.22
5	11.30 ± 0.74	3.98 ± 0.82	2.01 ± 0.22	1.86 ± 0.22

5.4 Graph

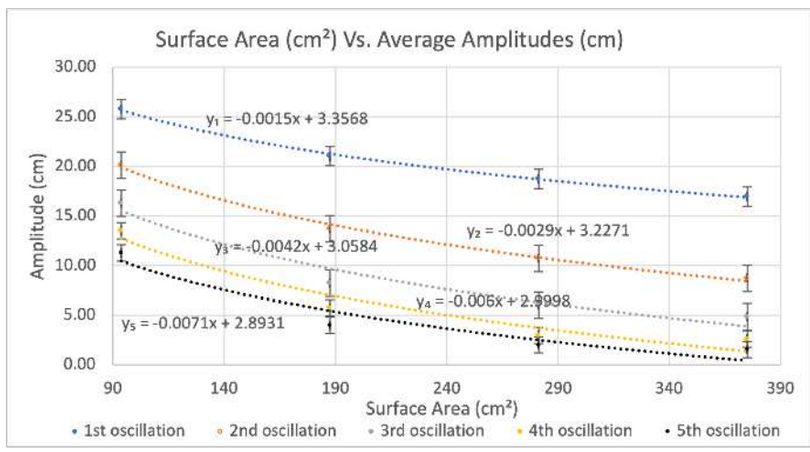


Figure 4: All of the Averages plotted against the number of periods, with vertical error bars being the largest uncertainty in its set. For example, ± 0.97 cm for the first oscillation.

As shown by the best-fit line, the model resembles an exponential function. The best-fit lines also pass within the error bars, suggesting that exponential equations fit the data well. I will confirm the results by linearizing the graph.

5.5 Linearized Graph

Clastify

Linearization entails taking the natural log of both sides of equation 5.

$$\ln x(t) = \ln\left(x_{max}\cos(\omega t)\right) - \frac{k}{2m}t$$

For the data set, I have to ln all the displacement values, so the y-axis becomes ln(*amplitude*). As for error bars, if I just ln my uncertainty, some values become negative, so I decided to use percentage error. Percentage error uses the largest uncertainty divided by its corresponding value, so from the 1st to the 5th oscillation respectively, we have

$$\frac{0.97}{21.04} = 4.61\%,\ \frac{1.34}{20.13} = 6.66\%,\ \frac{1.12}{8.41} = 13.3\%,\ \frac{0.82}{5.75} = 14.3\%,\ \frac{0.82}{3.98} = 20.6\%$$

number of oscillations	Averaged Displacement (cm)			
	93.75 cm²	187.5 cm²	281.25 cm²	375 cm²
1 (±4.61%)	3.25	3.05	2.93	2.83
2 (±6.66%)	3.00	2.62	2.37	2.17
3 (±13.3%)	2.79	2.11	1.80	1.58
4 (±14.3%)	2.60	1.75	1.09	0.96
5 (±20.6%)	2.42	1.38	0.70	0.44

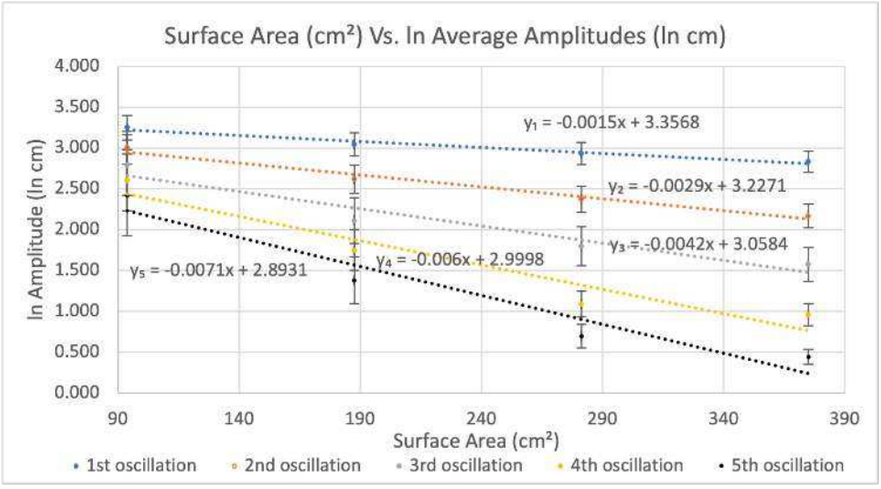


Figure 5: The lines presented are the best-fit lines calculated by Excel

The linear model fits well for the first three oscillations, indicating an exponential decay for the initial parts. However, it fails to model the last two oscillations, and I will discuss this further in the following section.

The linear model provides useful information. I will demonstrate by using the 1st oscillation as an example.

$$\text{Line of best-fit slope} = -0.0015$$

$$\text{y-intercept} = 3.3568$$

$$m_{max} = \frac{3.25 * (1 + 4.61\%) - 2.83 * (1 - 4.61\%)}{281.25} = -0.00249$$

$$m_{min} = \frac{3.25 * (1 - 4.61\%) - 2.83 * (1 + 4.61\%)}{281.25} = -0.000497$$

$$\text{Gradient Uncertainty} = |m_{max}| - |m_{min}| = 0.00199$$

$$\text{Slope}_1 = -0.0015 \pm 0.00199,\ \ -0.0015 \pm 132\%$$

Similar algebra derives

Clastify

$$\text{Slope}_2 = -0.0029 \pm 0.00245, \quad -0.0029 \pm 84.5\%$$

$$\text{Slope}_3 = -0.0042 \pm 0.00413, \quad -0.0042 \pm 119\%$$

$$\text{Slope}_4 = -0.006 \pm 0.00362, \quad -0.006 \pm 60.4\%$$

$$\text{Slope}_5 = -0.0071 \pm 0.00420, \quad -0.0071 \pm 59.2\%$$

6 Evaluation

6.1 Uncertainty

As seen in the gradient uncertainty, this experiment leaves a lot of room for improvement. However, the R^2 values demonstrated are also convincing enough to establish the exponential decay relationship between increasing surface area and amplitude on a pendulum.

As discussed in section 5.5, only the first three oscillations largely followed the exponential decay relationship. The most reasonable explanation is that air resistance was no longer the dominating damping factor in the last two periods, but was the friction between the string and the ring stand. The general drag equation is $\propto v^2$, as amplitude (x_{max}) decreases, velocity decreases because

$$v = \omega x_{max} \cos \omega t$$

Therefore, the effect of drag also decreases over time. However, for friction

$$F_f = \mu F_N, F_N = ma_{net}$$

Since mass never varies and acceleration from SHM is negligible compared to the gravitational acceleration, friction is constant throughout. Thus, after some time, it would have exceeded the drag force.

There are a few other sources of error.

- Lateral motion: Upon hand-held release, the mass inevitably undergoes some sideway movements. This additional movement causes the mass to experience more drag.
- The string: This analysis did not consider the additional mass and drag caused by the string, which would have contributed to the drag Vs. friction situation above because its drag decreases while friction remains constant.
- Random Error: In between changing the surface area from 187.5 cm^3 to 281.25 cm^3, I accidentally moved the camera. Although I tried my best to reconstruct the setup, there is a probability of systematic offset.

6.2 Improvements

Several improvements can be made to the experiment to lower uncertainty.

To address the friction problem, the string and the ring stand can be switched to smoother materials. Or, the experiment can be done in a more viscous fluid to increase the drag force since $F_D \propto \rho_{fluid}$.

The experiment can also benefit from more consistent mass. The paper mass here is connected by stapler clips, so the mass distribution was non-uniform, which could result in unnecessary lateral movements.

Furthermore, the measurement equipment can be more rigorous. Counting pixels can be replaced with a photogate timer, which is a laser sensor that records whenever the laser is blocked.

7 Conclusion

Clastify

The experiment demonstrates a negative logarithmic relationship between surface area and the extent of in simple harmonic motion, as seen in Figure 4; more specifically, as the area increases, the amplitude decreases. The reason behind such a phenomenon is air resistance, which is proportional to surface area. Thus, a larger area should contribute to more drag force.

However, as Figure 5 shows, this relationship stops holding after the 4th oscillation. Since drag drops over time because velocity decreases and friction between the string and the clamp is constant throughout, at some point, friction will exceed air resistance and become the dominant damping force.

For further investigations, the experiment can adopt a larger range of surface areas to improve the results of later oscillations, arriving at a more definitive conclusion. Moreover, the experiment can be repeated with more trials to reduce the statistical uncertainty.

The real-life application of this relationship applies to all simple harmonic motion, including pendulums, mass-spring systems, and 2D circular motion. For example, an outdoor suspicion system needs to account for the effect of wind to avoid losing control of the mass. Or, one of those amusement park rides with a giant swinging blob also needs to check air resistance when programming the system, or else the mechanical system cannot endure the long-term pressure. Therefore, understanding air resistance as a source of damping is crucial.